

End-to-end walk — OCP Access Tracking Dashboard 0.19.0 on CRC

Application 0.19.0 (commit 3629c8ca27), chart group-sync-dashboard-0.21.0, on the reference cluster OpenShift 4.18.2. Driven by Playwright (Chromium, 1440x900) through the real route <https://group-sync-dashboard.apps-crc.testing>, logged in as jane.smith through oauth-proxy and OpenShift OAuth. Run 2026-09-14T12:03:04Z → 2026-09-14T12:03:51Z. Every number below is read from the run's results files or measured on the downloaded artefacts; nothing is typed in from memory.

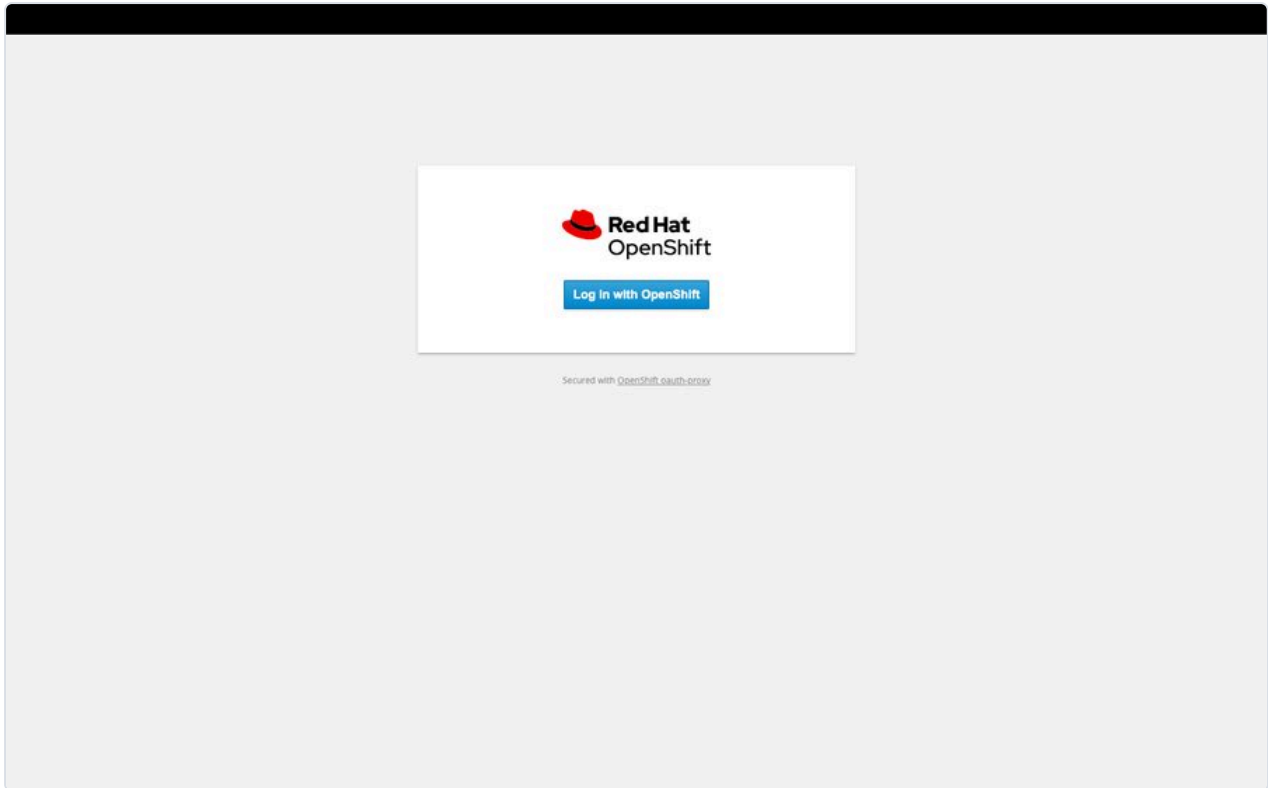
Outcome: PASS. 19 of 19 scripted steps passed in the main walk; 13 of 13 in the second pass. This walk is a **self-scope** reader (jane.smith), a non-administrator: reporting is for administrators only, so the Reports tab is refused and no report is generated. The value of this walk is what a narrowed reader is shown and withheld.

1. What was measured

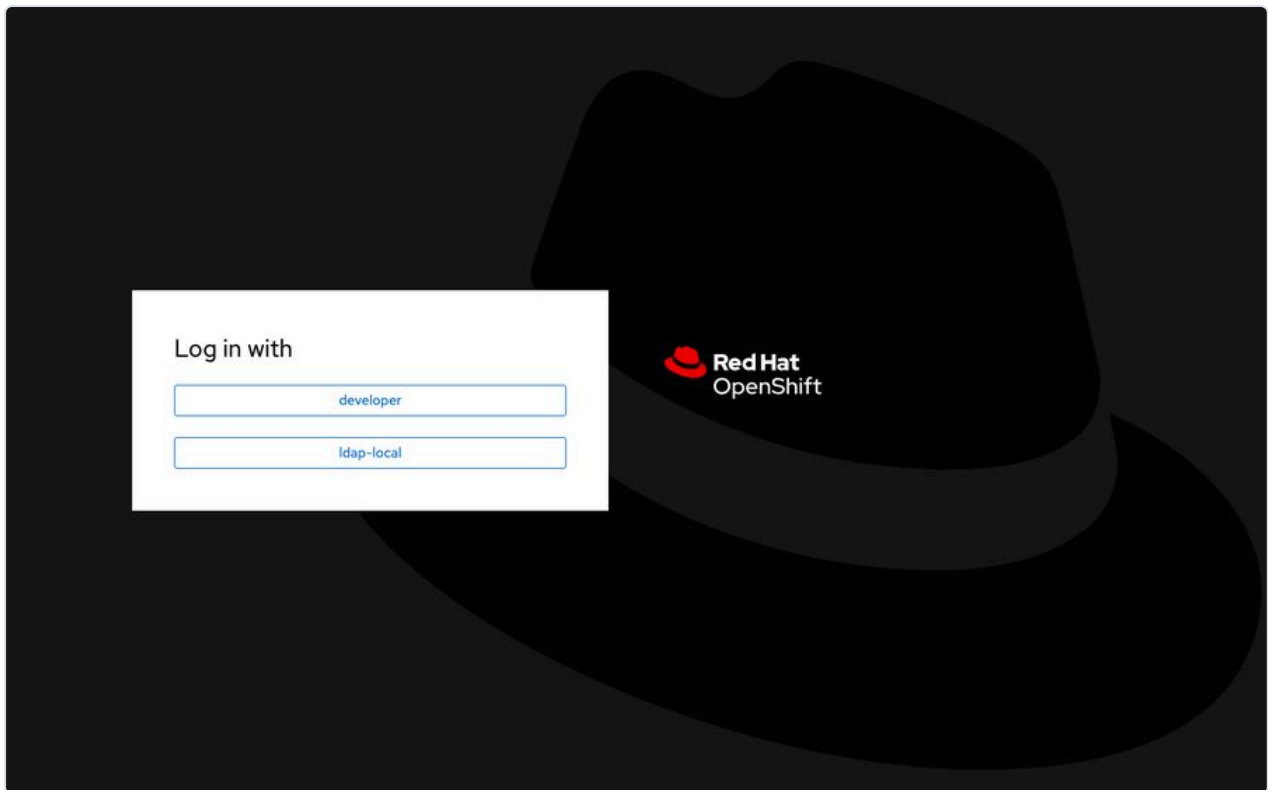
Measure	Value
Route	https://group-sync-dashboard.apps-crc.testing
Identity used	jane.smith (jane.smith@cluster.local), authenticated=True, scope self
Application / commit / branch	0.19.0 / 3629c8ca27 / feat/d2-per-cluster-authorization (dirty=False)
Features	export=True, reporting=True, reporting_prefix=/report
Chart / Helm revision	group-sync-dashboard-0.21.0 / revision 221 (2026-09-14 02:33:47)
Pods	group-sync-dashboard-5fb6c5b6d8-lg66b (2/2) group-sync-dashboard-report-cf59f655f-sntss (1/1)
Images	group-sync-dashboard: group-sync-dashboard:0.19.0-3629c8ca27; group-sync-dashboard-report: group-sync-dashboard-report:0.19.0-3629c8ca27;
Tabs offered	9 tabs: Overview, Groups, Users, Access granted, RBAC policy, Namespace audit, Logins, Usage, Reports
Reader scope	self — reporting refused (non-administrator)
Scripted steps	main walk 19 passed / 0 failed; second pass 13 / 13; PDF first pages rendered 0 / 0

2. Login through the real ingress

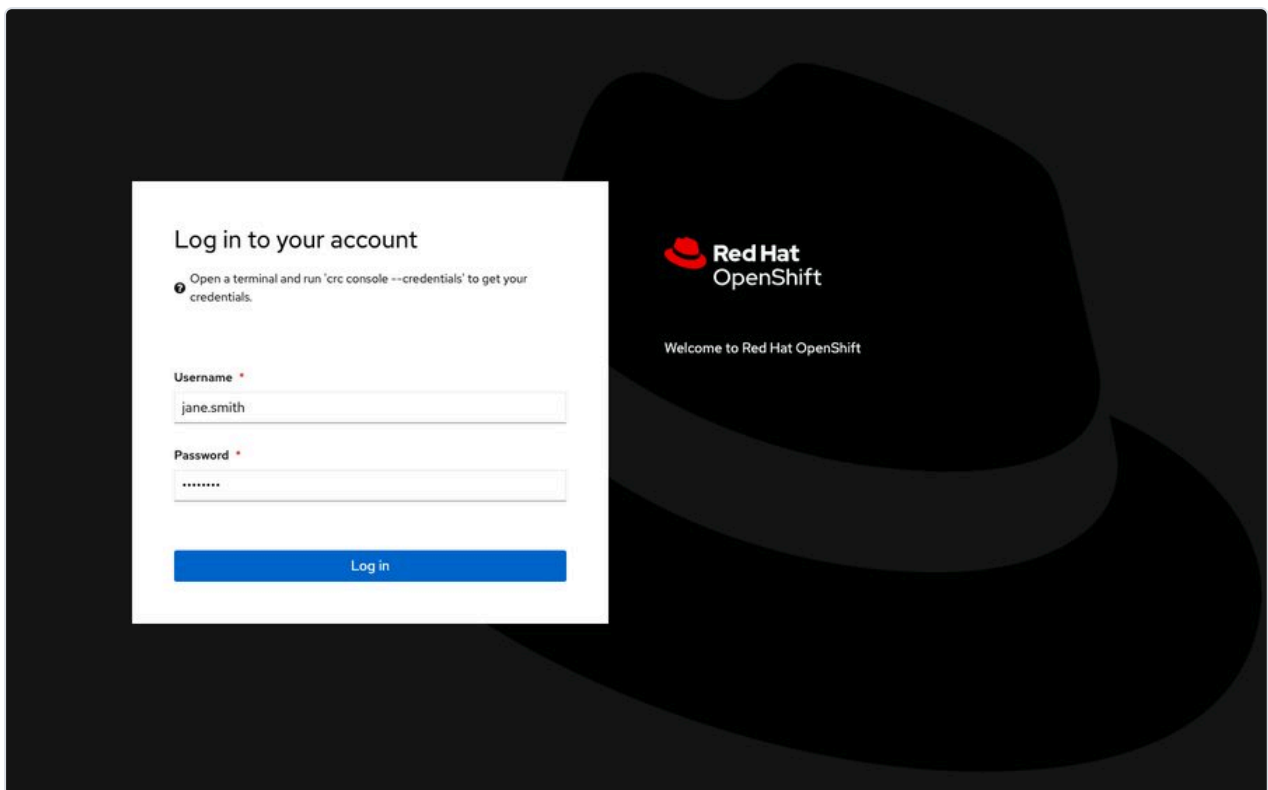
oauth-proxy's interstitial, OpenShift's identity-provider chooser (two providers on this lab), the credential form, then the dashboard. The ServiceAccount is the OAuth client, so a consent page can appear; this session did not show one.



open the route — <https://group-sync-dashboard.apps-crc.testing> → <https://group-sync-dashboard.apps-crc.testing/>



identity-provider chooser — offered: developer, ldap-local; picked ldap-local



OpenShift login form — username filled: jane.smith

✓ dashboard rendered at <https://group-sync-dashboard.apps-crc.testing/#page=overview&cluster=crc-local>

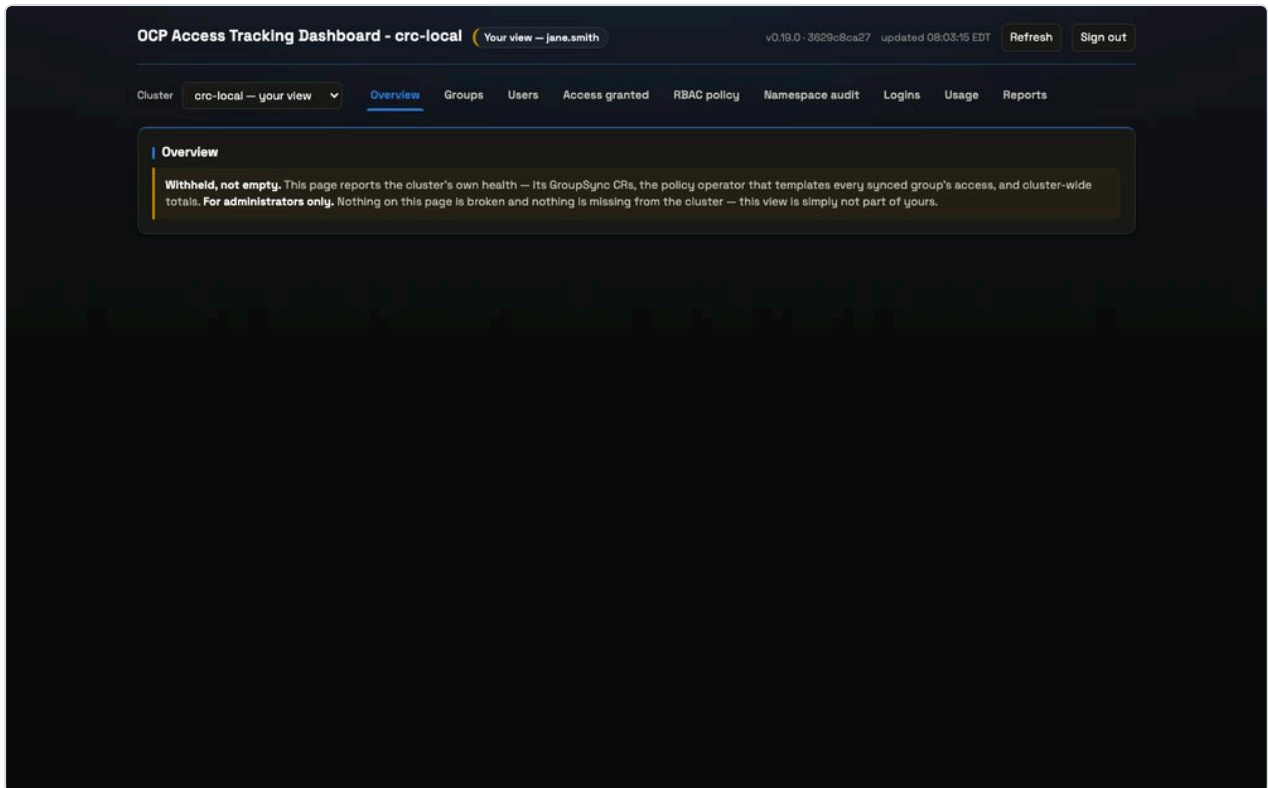
3. API probes from inside the session

Same-origin fetches issued by the page, so the proxy cookie travels with them.

Endpoint	HTTP	Body (abridged)
/api/whoami	HTTP 200	<pre>{"user":"jane.smith","email":"jane.smith@cluster.local","authenticated":true,"logout_url":"/oauth/sign_out","session":{"cookie_expire_seconds":14400,"cookie_refresh_seconds":0,"idle_timeout":{"enabled":false}},"visibility":{"scope":"self","enabled":true,"clusters":{"crc-local":{"policy":"inherit","identity":"same-as-host","scope":"self"}}}}</pre>
/api/version	HTTP 200	<pre>{"leader":true,"version":"0.19.0","commit":"3629c8ca27","branch":"feat/d2-per-cluster-authorization","dirty":false,"features":{"export":true,"reporting":true,"reporting_prefix":"/report"},"timezone":{"name":"America/New_York","abbrev":"EDT","utc_offset":"-0400"}}</pre>
/api/clusters	HTTP 200	<pre>[{"id":"crc-local","enabled":true,"reachable":true,"status":"ok","last_poll":"2026-09-14T12:03:05Z","group_count":66,"groupsync_count":3,"visibility":{"policy":"inherit","scope":"self"}},{"id":"prod-east","enabled":true,"reachable":true,"status":"ok","last_poll":"2026-09-14T07:33:36Z","group_count":66,"groupsync_count":3,"visibility":{"policy":"inherit","scope":"self"}}]</pre>
/report/healthz	HTTP 200	<pre>{"status":"ok"}</pre>
/report/readyz	HTTP 200	<pre>{"status":"ready","snapshot":"2026-09-14T12:00:08.441212Z","schema":11}</pre>

4. Every tab

Each tab is clicked, the page waits for the tab to report `aria-current`, the network to settle and a heading to exist, and the capture is refused if any JavaScript error or a visible “Dashboard API error” occurred.



tab Overview — heading: Overview

OCP Access Tracking Dashboard - crc-local

Your view — jane.smith

v0.19.0 · 3629c8ca27 updated 08:03:19 EDT

Refresh

Sign out

Cluster

crc-local — your view

Find

filter by name

Group state

all

Export

CSV

JSON

17 rows

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Groups · 17 shown

Showing only the groups jane.smith belongs to. The cluster-wide list is the administrator tier.

NAME	MEMBERS	OWNER	LAST REFRESHED	SOURCE DN
app-ocp-rbac-alpha-cluster-admin	1	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-alpha-cluster-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-alpha-cluster-developer	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-alpha-cluster-developer,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-alpha-ns-admin	1	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-alpha-ns-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-alpha-ns-developer	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-alpha-ns-developer,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-beta-ns-admin	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-beta-ns-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-demo-cluster-developer	1	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-demo-cluster-developer,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-demo-ns-admin	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-demo-ns-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-demo-ns-developer	3	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-demo-ns-developer,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-platform-cluster-developer	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-platform-cluster-developer,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-platform-ns-admin	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-platform-ns-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-spar-ns-admin	1	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-spar-ns-admin,ou=Groups,dc=ephico2real,dc=com
app-ocp-rbac-spar-ns-developer	2	app-ocp-rbac-group-groupsync	3m ago	cn=app-ocp-rbac-spar-ns-developer,ou=Groups,dc=ephico2real,dc=com
app-ssb-autobahnusers	8	autobahnusers-onboarding-groupsync	3m ago	cn=app-ssb-autobahnusers,ou=Groups,dc=ephico2real,dc=com
bda-rbac-presto-alpha-users	1	bdp-oud-group-rbac-groupsync	2m ago	cn=bda-rbac-presto-alpha-users,ou=Groups,dc=ephico2real,dc=com
bda-rbac-spark-alpha-users	2	bdp-oud-group-rbac-groupsync	3m ago	cn=bda-rbac-spark-alpha-users,ou=Groups,dc=ephico2real,dc=com
bda-rbac-trino-alpha-apps	1	bdp-oud-group-rbac-groupsync	2m ago	cn=bda-rbac-trino-alpha-apps,ou=Groups,dc=ephico2real,dc=com
bda-rbac-trino-delta-apps	1	bdp-oud-group-rbac-groupsync	2m ago	cn=bda-rbac-trino-delta-apps,ou=Groups,dc=ephico2real,dc=com

Select a group to see its members.

tab Groups — heading: Groups · 17 shown

OCP Access Tracking Dashboard - crc-local

Your view — jane.smith

v0.19.0 · 3629c8ca27 updated 08:03:19 EDT

Refresh

Sign out

Cluster

crc-local — your view

Find

filter by id or name

Export

CSV

JSON

1 row

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Users · 1 shown

Showing only your own row, jane.smith. The cluster-wide user list is the administrator tier.

Users read from the cluster 2026-09-14 08:03:05 EDT (13s ago).

USER	STATUS	PROVIDER	GROUPS	LAST LOGIN	FIRST SEEN IN A GROUP
jane.smith > Jane Smith	logged in since 2026-08-05 12:14:14 EDT (39d ago) (approx.)	ldap-local	17	2026-09-05 22:19:25 EDT (8d ago)	2026-08-02 00:00:33 EDT (43d ago)

Select a user to see every group they belong to and every binding that reaches them.

tab Users — heading: Users · 1 shown

OCP Access Tracking Dashboard - crc-local

Your view — jane.smith

v0.19.0 · 3629c8ca27 · updated 08:03:19 EDT

Refresh

Sign out

Cluster

crc-local — your view

Export

CSV

JSON

10 rows

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Access granted

Showing the access that reaches **jane.smith** through your groups. The cluster-wide list — every group's grants — is the administrator tier.

Access granted · direct role bindings

These bindings reach you through the group in the Via group column. Select the group to see who else is in it. Role rules are **not** evaluated, so this is what has been granted, not a computed list of permitted actions.

Namespaces

Cluster-wide roles

4

5

Cluster-wide

ROLE	BINDING	VIA GROUP
admin ClusterRole	app-ocp-rbac-alpha-cluster-admin-crb	app-ocp-rbac-alpha-cluster-admin >
edit ClusterRole	app-ocp-rbac-alpha-cluster-developer-crb	app-ocp-rbac-alpha-cluster-developer >
edit ClusterRole	app-ocp-rbac-demo-cluster-developer-crb	app-ocp-rbac-demo-cluster-developer >
edit ClusterRole	app-ocp-rbac-platform-cluster-developer-crb	app-ocp-rbac-platform-cluster-developer >
edit ClusterRole	demo-cluster-developer-crb	app-ocp-rbac-demo-cluster-developer >

Namespaced

NAMESPACE	ROLE	BINDING	VIA GROUP
demo-qa	edit ClusterRole	demo-developer-rb	app-ocp-rbac-demo-ns-developer >
demo-rnd	edit ClusterRole	demo-developer-rb	app-ocp-rbac-demo-ns-developer >
demo-uat	edit ClusterRole	demo-developer-rb	app-ocp-rbac-demo-ns-developer >
openshift-user-workload-monitoring	monitoring-rules-edit ClusterRole	demo-monitoring-config-rb	app-ocp-rbac-demo-ns-admin >
openshift-user-workload-monitoring	monitoring-rules-edit ClusterRole	demo-monitoring-config-rb	app-ocp-rbac-demo-ns-developer >

tab Access granted — heading: Access granted

OCP Access Tracking Dashboard - crc-local

Your view — jane.smith

v0.19.0 · 3629c8ca27 · updated 08:03:20 EDT

Refresh

Sign out

Cluster

crc-local — your view

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

RBAC policy

Withheld, not empty. The grants that bypass the policy system name other people's access. **For administrators only.** Nothing on this page is broken and nothing is missing from the cluster — this view is simply not part of yours.

tab RBAC policy — heading: RBAC policy

OCP Access Tracking Dashboard - crc-local

Your view — jane.smith

v0.19.0 · 3629c8ca27 · updated 08:03:21 EDT

Refresh

Sign out

Cluster

crc-local — your view

Export

CSV

JSON

1 row

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Namespace audit

Showing only roles granted directly to jane.smith. The per-namespace audit of everyone's grants is the administrator tier.

GRANTS	SCOPE	BINDING
user-settings-56e5d658-1cee-4dad-832e-b56477b707b4-role Role	openshift-console-user-settings	user-settings-56e5d658-1cee-4dad-832e-b56477b707b4-rolebinding

A direct grant survives offboarding. Consider asking for the equivalent group membership and having the binding removed — the runbook on the administrator's view of this tab is the same one that applies to yours.

tab Namespace audit — heading: Namespace audit

Cluster

crc-local — your view

Outcome

all outcomes

Export

CSV

JSON

32 rows · partial: 32 of 53 loaded

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Login attempts

Showing only sign-in attempts recorded for **jane.smith**. Attempts are recorded under the name as typed, so an attempt typed with different capitalisation is a different name and is not shown here.

53 sign-in attempts recorded for you

Most recent
8d ago

Watching since 2026-08-07 19:47:59 EDT (37d ago); oldest attempt still retained 2026-09-06 10:37:34 EDT; last read 2026-09-14 08:03:06 EDT (17s ago). Nothing before capture began was ever recorded — the oauth-server's log dies with its pod and cannot be read backwards — so an empty list here means nothing was observed, never that nobody signed in. The oldest attempt predates the first read because that read looked back an hour, which is expected rather than a fault.

Your cluster access

in the login gate group

Membership of is what allows you to sign in to this cluster. Who else is or is not in the gate group is the administrator tier.

Your attempts						
WHEN	USER	OUTCOME	PROVIDER	DETAIL	NODE	
2026-09-05 22:19:24 EDT	jane.smith > · Jane Smith	signed in	ldap-local	audit: credential allow via /login/dap-local	crc	
2026-09-05 15:42:43 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-09-04 10:10:21 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-09-03 18:42:20 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-09-03 18:15:44 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-09-03 12:13:49 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-12 13:12:35 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-11 20:36:34 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-11 18:33:03 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-11 16:58:28 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-11 12:53:15 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-11 00:24:07 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-10 21:09:28 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-10 09:01:37 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-10 08:49:49 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-09 19:45:50 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-09 17:54:34 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-09 01:03:37 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-08 16:16:38 EDT	jane.smith > · Jane Smith	wrong password	ldap-local	LDAP result code 49 (LDAP 49)	66444df7fc-nccmh	
2026-08-08 16:15:48 EDT	jane.smith > · Jane Smith	wrong password	ldap-local	LDAP result code 49 (LDAP 49)	66444df7fc-nccmh	
2026-08-07 20:02:54 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-07 20:02:46 EDT	jane.smith > · Jane Smith	wrong password	ldap-local	LDAP result code 49 (LDAP 49)	66444df7fc-nccmh	
2026-08-07 20:01:51 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-07 19:49:02 EDT	jane.smith > · Jane Smith	failed — no reason given	ldap-local	audit: cli deny via openshift-challenging-client — Authentication failed, attempted: basic	crc	
2026-08-07 19:49:01 EDT	jane.smith > · Jane Smith	wrong password	ldap-local	LDAP result code 49 (LDAP 49)	66444df7fc-nccmh	
2026-08-07 19:48:58 EDT	jane.smith > · Jane Smith	signed in	ldap-local	audit: cli allow via openshift-challenging-client	crc	
2026-08-07 19:48:57 EDT	jane.smith > · Jane Smith	signed in	ldap-local		66444df7fc-nccmh	
2026-08-07 12:56:29 EDT	jane.smith > · Jane Smith	failed — no reason given	ldap-local	audit: cli deny via openshift-challenging-client — Authentication failed, attempted: basic	crc	
2026-08-07 12:26:37 EDT	jane.smith > · Jane Smith	signed in	ldap-local	audit: cli allow via openshift-challenging-client	crc	
2026-08-07 12:15:28 EDT	jane.smith > · Jane Smith	failed — no reason given	ldap-local	audit: cli deny via openshift-challenging-client — Authentication failed, attempted: basic	crc	
2026-08-05 12:15:48 EDT	jane.smith > · Jane Smith	signed in	ldap-local	audit: cli allow via openshift-challenging-client	crc	
2026-08-05 12:14:15 EDT	jane.smith > · Jane Smith	signed in	ldap-local	audit: cli allow via openshift-challenging-client	crc	

No match covers two cases: this tool cannot generate a real membership in set in the login gate group, and a username that does not exist. The identity provider's search filter denies the rest.

group, so both produce the same `no entries matching` line — telling them apart from the log alone would need a directory read this dashboard does not make. Where a **second badge** appears beside it, that ambiguity **HAS** been resolved, from synced group membership rather than from any new read: hover it for the reasoning. Where no second badge appears, no login gate is known and the honest answer is still that we cannot tell. **Detail** is the directory's own diagnostic, passed through rather than interpreted. On Active Directory it carries a `data <hex>` sub-code, which is what distinguishes an expired password from a locked account — the LDAP result code alone does not: 49 is returned for all of them. OpenLDAP leaves the field empty and signals expiry with code 53 instead. **Replica** is which `oauth-server` pod saw it, with the common prefix trimmed; the two replicas serve different attempts, so a name here is the pod whose log to read.

tab Logins — heading: Login attempts

OCP Access Tracking Dashboard

Your view — jane.smith

v0.19.0 · 3629c8ca27 · updated 08:03:26 EDT

Refresh

Sign out

Overview

Groups

Users

Access granted

RBAC policy

Namespace audit

Logins

Usage

Reports

Dashboard usage

Showing jane.smith activity only. This is identifiable personnel data, so each person sees their own.

0

no activity today

Days recorded

Interactions

Retention

8

208

400 days

By user and day

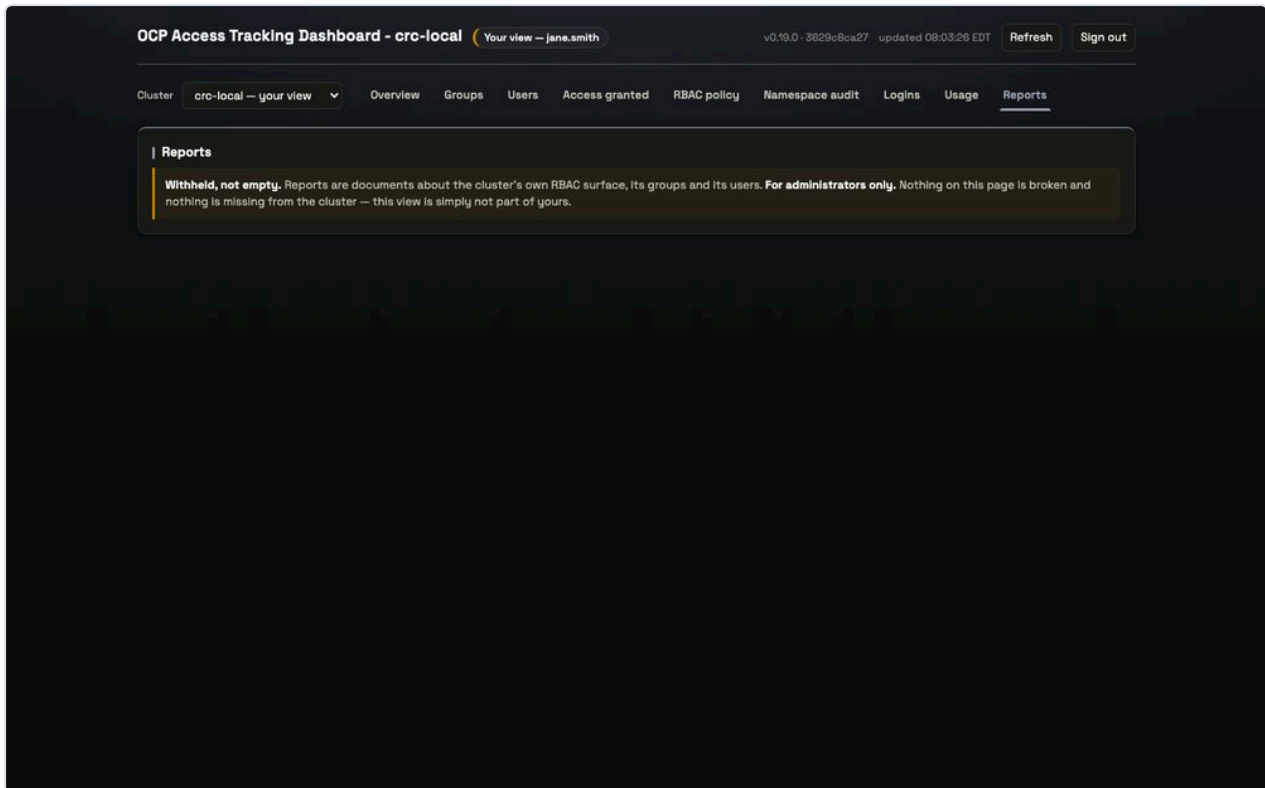
USER	DAY (UTC)	FIRST SEEN	LAST SEEN	INTERACTIONS
jane.smith · jane.smith@cluster.local	2026-09-06	22:19:27 EDT	22:19:49 EDT	6
jane.smith · jane.smith@cluster.local	2026-09-05	15:42:46 EDT	15:43:02 EDT	11
jane.smith · jane.smith@cluster.local	2026-09-04	10:10:23 EDT	10:11:32 EDT	6
jane.smith · jane.smith@cluster.local	2026-09-03	12:13:50 EDT	18:50:07 EDT	60
jane.smith · jane.smith@cluster.local	2026-08-12	20:36:03 EDT	13:12:45 EDT	12
jane.smith · jane.smith@cluster.local	2026-08-11	21:09:32 EDT	18:38:06 EDT	60
jane.smith · jane.smith@cluster.local	2026-08-10	08:49:53 EDT	09:02:05 EDT	17
jane.smith · jane.smith@cluster.local	2026-08-09	01:03:45 EDT	19:48:13 EDT	36

Times render in the server's timezone; the **Day** column is a UTC date. That split is deliberate rather than an oversight: the **day** is a stored bucket, computed server-side when the activity was recorded, so re-labelling it local would misstate rows already written, and re-bucketing would make the same column mean one thing before a deploy and another after. Near midnight a session can therefore sit on the following UTC day. An interaction is one deliberate action — opening the dashboard, switching tab, drilling in, changing a filter — not one HTTP request: the page refreshes itself every 60s and counting those measured how long a tab was left open rather than whether anyone used it. Reading time is invisible, so a long look at one page counts once. These are not logins either: the proxy owns this dashboard's session, so nothing here records a sign-in to it. Logins to the cluster are a separate record with a separate source — the `oauth-server`'s own log — and they are on the **Logins** tab.

tab Usage — heading: Dashboard usage

5. Reports — refused for this reader

reporting is refused for this reader: Withheld, not empty. Reports are documents about the cluster's own RBAC surface, its groups and its users. For administrators only. Nothing on this page is broken and nothing is missing from the clust

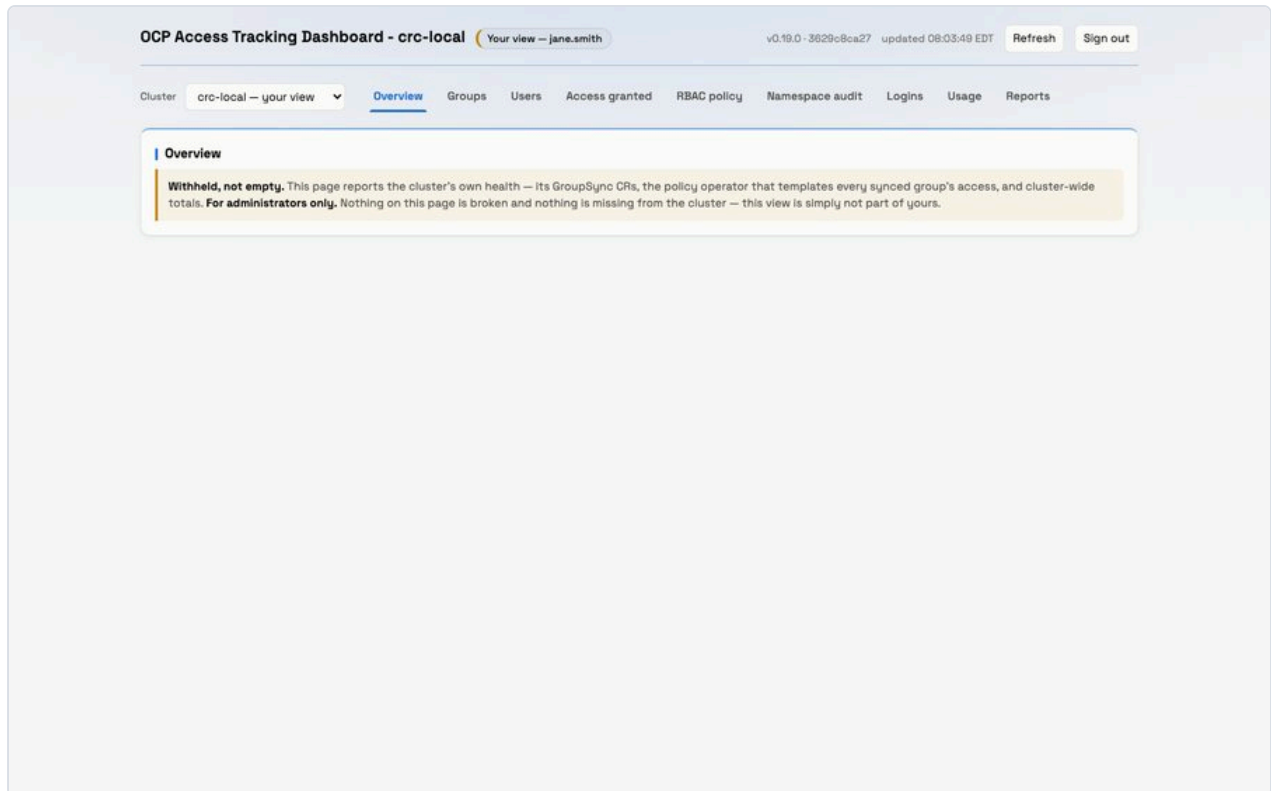


The Reports tab as a non-administrator: withheld, not empty

A report is a document about the cluster's whole RBAC surface — every group, every user, every binding — which cannot be scoped to one reader's own access the way a membership list can. So the dashboard refuses to mint a report ticket for a non-administrator (a 403 at `/api/report/ticket` with the sentence “For administrators only”), and the Reports tab shows the refusal rather than a broken or empty page. No report was generated; §6 and §7 do not apply to this walk.

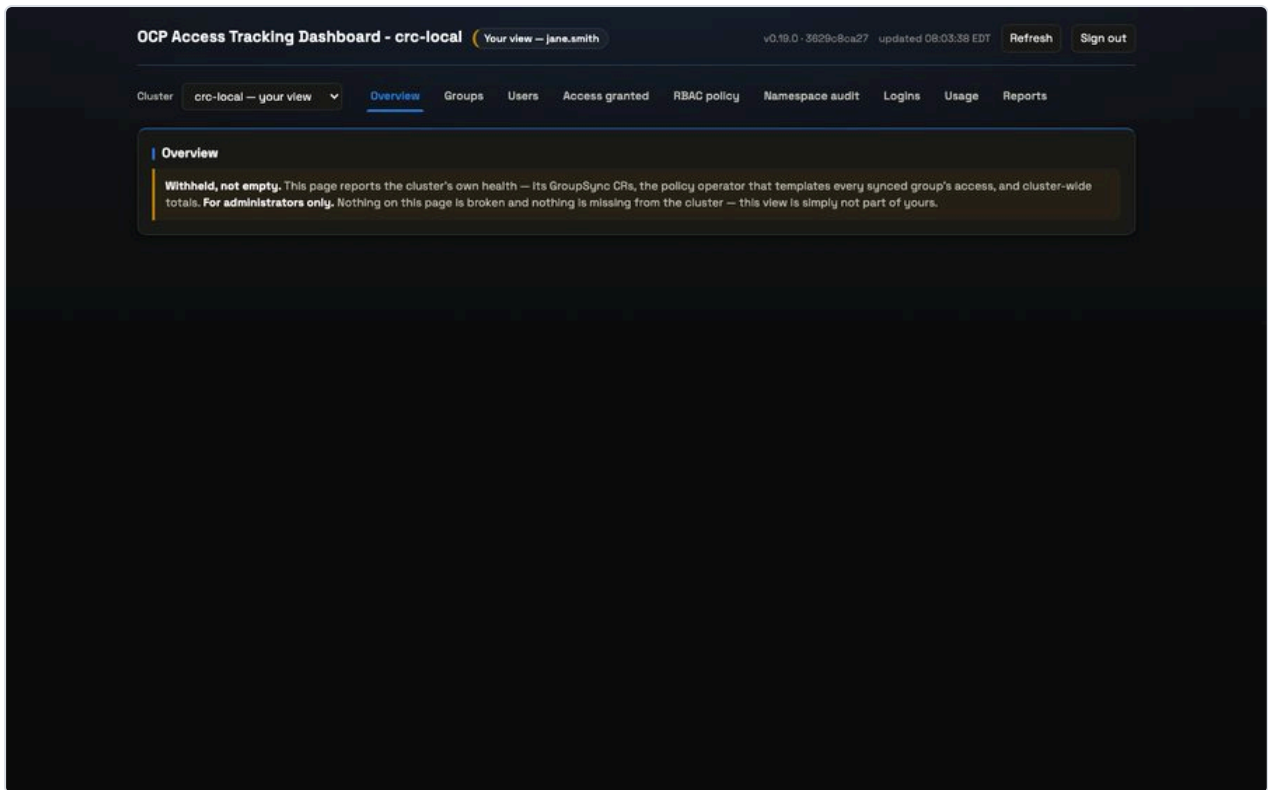
8. Second cluster, light theme

The cluster selector offers: crc-local — your view, prod-east — your view. Scope pill: “Your view — jane.smith”.

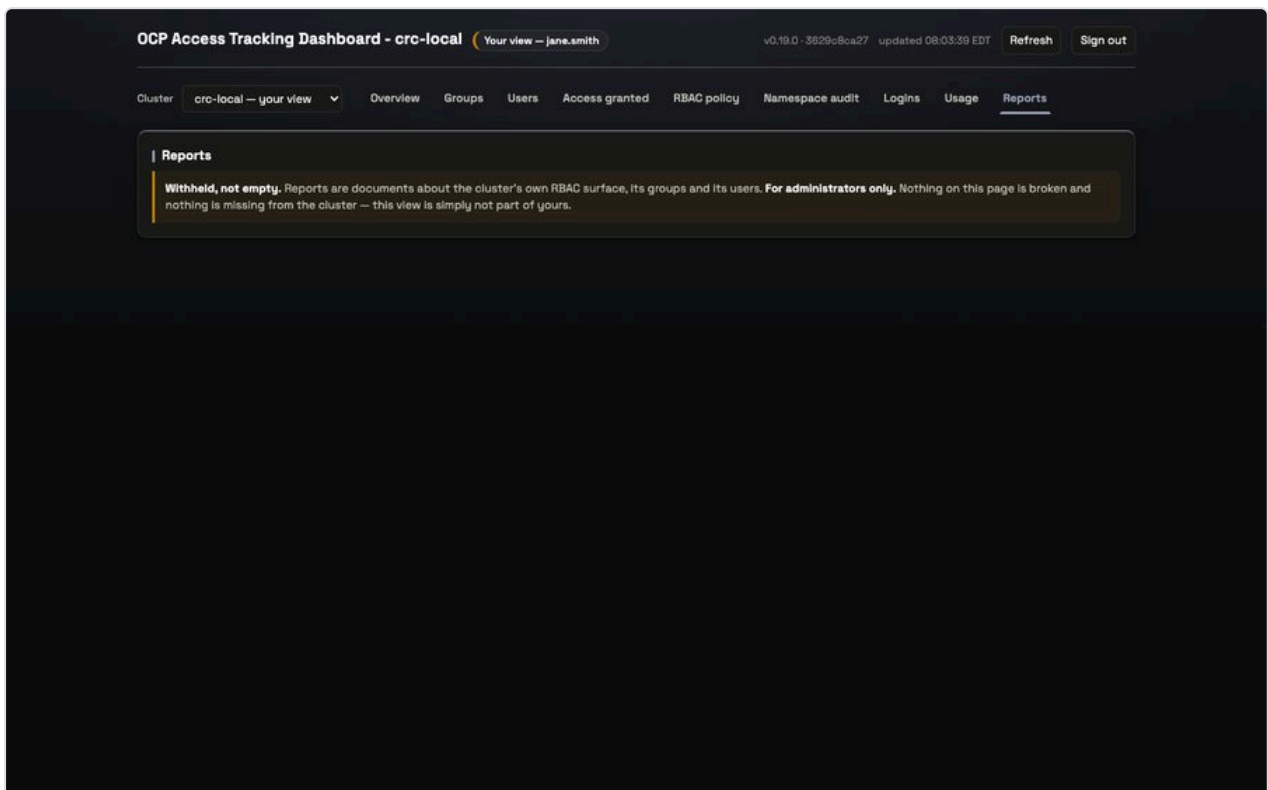


Overview in the light theme — prefers-color-scheme: light

✓ switch cluster to crc-local — your view — heading: Overview; url <https://group-sync-dashboard.apps-crc.testing/#page=overview&cluster=crc-local>



switch cluster to crc-local — your view — heading: Overview; url <https://group-sync-dashboard.apps-crc.testing/#page=overview&cluster=crc-local>



Reports tab on crc-local — your view — no report form; the card says: Skip to main content OCP Access Tracking Dashboard - crc-local Your view — jane.smith v0.19.0 · 3629c8ca27 updated 08:03:39 EDT Refresh Sign out Cluster crc-local — your view prod-east — your view Overview Groups Users Access granted RBAC policy Namespace audit Logins Usage Reports Reports Withheld,

9. Findings

Finding 1 — the removed cluster prod-east is still offered and 404s when selected (issue #96). The self-tier walk hit the same lingering cluster the admin walk did: selecting prod-east renders "Dashboard API error: 404 Not Found on /api/clusters/prod-east/groupsyncs". prod-east was removed from the configuration during the D2 live check; the live ConfigMap lists only crc-local. Filed 2026-09-14 as #96.

Finding 2 — the report pod returned 502 for a wide-tier reader mid-walk (issue #97). An earlier admin-tier walk this session caught the Reports tab showing "Dashboard API error: 502 Bad Gateway on /report/api/reports": the report pod had just been preempted on the saturated CRC node and was restarting. A self-tier reader like jane.smith never calls the report service (the refusal is client-side), so this did not affect the jane.smith walk. Filed 2026-09-14 as #97 (expose priorityClassName).

10. Files delivered with this document

File	Bytes	What
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Screenshots: 12 in the main walk and 9 in the second pass, at 1440×900 device pixels (full-page where the tab scrolls); the images in this document are downscaled copies.